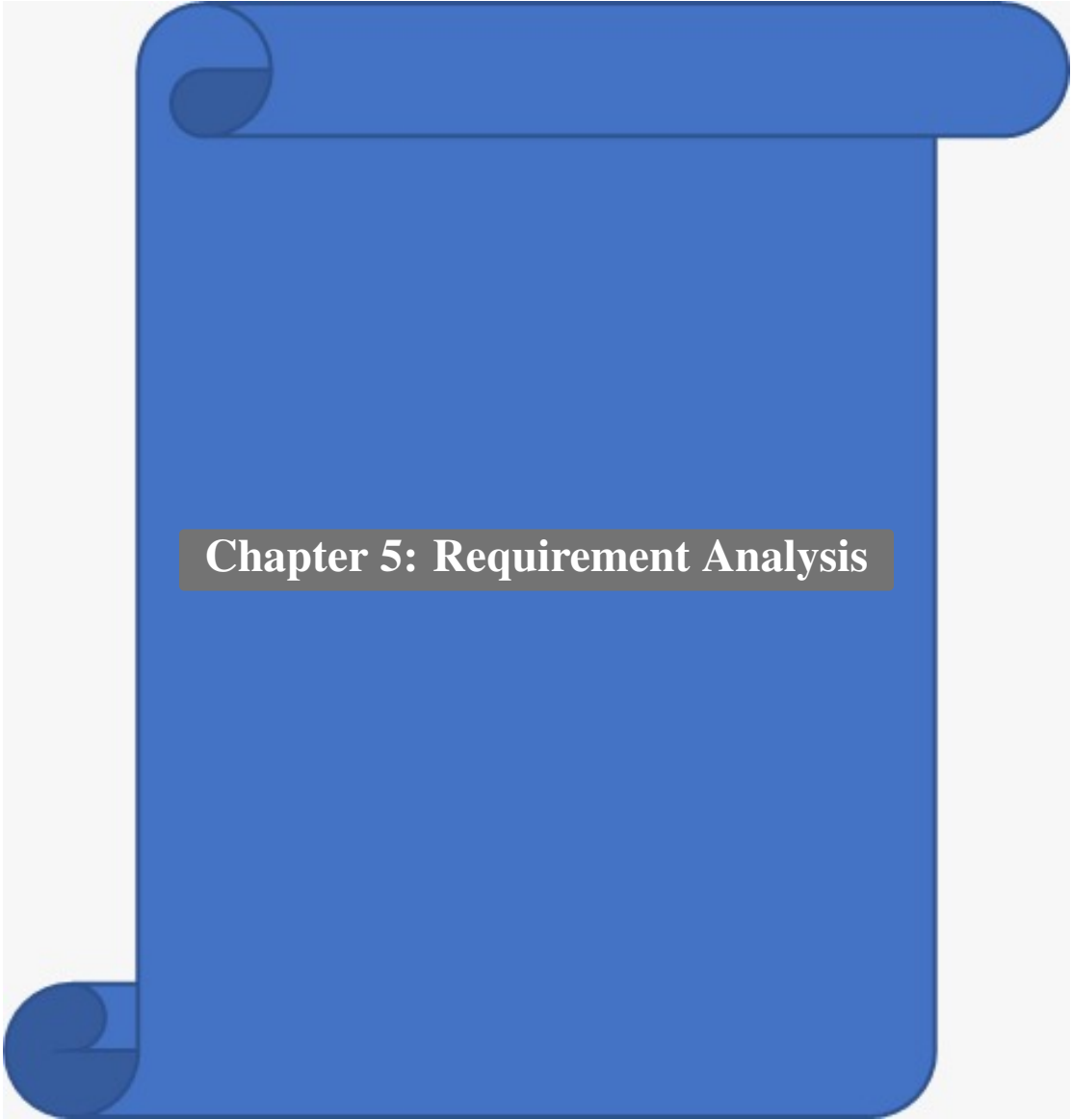




Chapter 5: Requirement Analysis

5.1 Hardware Requirements

Table 5.1: Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above / AMD Ryzen 3 or above
RAM	4 GB minimum (8 GB recommended)
Storage	256 GB SSD (minimum)
Display	1366 × 768 resolution (minimum)
Network	Broadband internet connection
Client Device	Any device with a modern web browser (Desktop, Tablet, Smartphone)

5.2 Software Requirements

Table 5.2: Software Requirements

Component	Specification
Operating System	Windows 10+, macOS 12+, or Linux (Ubuntu 20.04+)
Runtime	Node.js 18+ with npm
Framework	Next.js 15.5 with App Router
Language	TypeScript 5.0
Database	Firebase Firestore (NoSQL, Cloud-hosted)
Authentication	Firebase Authentication
File Storage	Firebase Storage
Payment Gateway	Razorpay API
AI Services	Google Genkit AI
Styling	Tailwind CSS 3.0, Radix UI
Build Tool	Turbopack
Browser	Google Chrome 90+, Firefox 88+, Safari 14+, or Edge 90+
Version Control	Git with GitHub
Deployment	Firebase Hosting / Vercel

5.3 Functional Requirements

The following table lists the functional requirements of the AgriTrace platform:

Table 5.3: Functional Requirements

ID	Requirement	Description	Priority
FR-01	User Registration	System shall allow new users to register with email and password, selecting their role (Farmer, Agent, Admin)	High
FR-02	User Authentication	System shall authenticate users via Firebase Authentication with email/password credentials	High
FR-03	Password Reset	System shall allow users to reset their password via email verification link	Medium
FR-04	Waste Listing Creation	Farmers shall be able to create waste listings with title, category, quantity, price, location, photos, and description	High
FR-05	Listing Discovery	Agents shall be able to browse all open waste listings sorted by creation date	High
FR-06	Listing Claiming	Agents shall be able to claim open listings, changing status to ASSIGNED	High
FR-07	Pickup Workflow	Agents shall transition listings through ASSIGNED → IN_TRANSIT → DELIVERED states	High
FR-08	Listing Cancellation	Farmers shall be able to cancel their own OPEN listings	Medium
FR-09	Payment Processing	System shall integrate Razorpay for processing payments between buyers and sellers	High
FR-10	Carbon Credit Calculation	System shall calculate carbon offsets based on waste type and quantity using established emission factors	Medium
FR-11	Rewards System	System shall track user points, streaks, and milestones based on platform activity	Low
FR-12	Photo Upload	Users shall be able to upload waste photos via Cloudinary integration during listing creation	Medium

ID	Requirement	Description	Priority
FR-13	Location Services	System shall capture GPS coordinates and display map-based location picker for waste reports	Medium
FR-14	Notifications	System shall send in-app notifications for status changes, assignments, and system events	Medium
FR-15	Admin Dashboard	Administrators shall view platform-wide statistics including user counts, listing counts by status, and activity trends	High
FR-16	CSV Export	Farmers shall be able to export their listing data as CSV files for offline record-keeping	Low
FR-17	User Role Management	Administrators shall be able to view all users and edit user roles	High
FR-18	Real-time Updates	All data changes shall be reflected in real-time across all connected clients using Firestore onSnapshot listeners	High

5.4 Non-Functional Requirements

The following table lists the non-functional requirements:

Table 5.4: Non-Functional Requirements

ID	Requirement	Description	Priority
NF-01	Performance	Pages shall load within 3 seconds on a standard broadband connection	High
NF-02	Scalability	System shall support up to 10,000 concurrent users without degradation	Medium
NF-03	Security	All data transmission shall be encrypted using HTTPS/TLS. Firebase security rules shall enforce role-based access control	High
NF-04	Availability	System shall maintain 99.5% uptime, leveraging Firebase's SLA guarantees	High

ID	Requirement	Description	Priority
NF-05	Responsiveness	UI shall be fully responsive across desktop (1920×1080), tablet (768×1024), and mobile (375×667) viewports	High
NF-06	Usability	Interface shall follow consistent design patterns with clear navigation, feedback messages, and loading indicators	Medium
NF-07	Maintainability	Codebase shall use TypeScript strict mode, consistent naming conventions, and modular file organization	Medium
NF-08	Data Integrity	Firestore transactions shall ensure atomicity for multi-document operations (e.g., order creation)	High
NF-09	Browser Compatibility	Application shall function correctly on Chrome 90+, Firefox 88+, Safari 14+, and Edge 90+	Medium
NF-10	Accessibility	All interactive elements shall have proper ARIA labels and keyboard navigation support via Radix UI primitives	Low